fixed and unchanging. You might ask whether we really know this to be true. In fact, we don't. But our success in describing numerous features of the universe, from a brief moment after the big bang right through to the present, assures us that if the laws are changing they must be doing so very slowly. The simplest assumption that is consistent with all that we know is that the laws are fixed.

Now imagine a universe in which the laws of physics are as parochial as local culture—changing unpredictably from place to place and defiantly resisting any outside influence to conform. Like the adventures of Gulliver, travels in such a world would expose you to an enormously rich array of unpredictable experiences. But from a physicist's perspective, this is yet another nightmare. It's hard enough, for instance, to live with the fact that laws that are valid in one country—or even one state—may not be valid in another. But imagine what things would be like if the laws of nature were as varied. In such a world experiments carried out in one locale would have no bearing on the physical laws relevant somewhere else. Instead, physicists would have to redo experiments over and over again in different locations to probe the local laws of nature that hold in each. Thankfully, everything we know points toward the laws of physics being the same everywhere. All experiments the world over converge on the same set of underlying physical explanations. Moreover, our ability to explain a vast number of astrophysical observations of far-flung regions of the cosmos using one, fixed set of physical principles leads us to believe that the same laws do hold true everywhere. Having never traveled to the opposite end of the universe, we can't definitively rule out the possibility that a whole new kind of physics prevails elsewhere, but everything points to the contrary.

Again, this does not mean that the universe looks the same—or has the same detailed properties—in different locations. An astronaut jumping on a pogo stick on the moon can do all sorts of things that are impossible to do on earth. But we recognize that the difference arises because the moon is far less massive than the earth; it does not mean that the law of gravity is somehow changing from place to place. Newton's, or more precisely, Einstein's, law of gravity is the same on earth as it is on the moon. The difference in the astronaut's experience is one of change in environmental detail, not variation of physical law.